

Testing Report — KNMP VisitHub

Document Title: Testing Report **System:** KNMP VisitHub — A Web-Based Booking and Visit Planning System for the Kwame Nkrumah Memorial Park **Author:** Manus AI **Date:** August 2026 **Revision:** 1.0

1. Introduction

This report summarises the testing approach, the automated test inventory, and the results achieved for KNMP VisitHub at the completion of the seventh enhancement round. All results quoted below were produced against the live project build and its TiDB Serverless database.

2. Test Strategy

Testing followed a layered strategy built on **Vitest 2.1** with mocked database modules for logic tests and selective live-database tests for the query helpers where provider-specific behaviour mattered.

Layer	Scope	Technique
Unit / integration (mocked DB)	Business rules, procedure permissions, email flow	Vitest with <code>vi.mock</code> over <code>./db</code> and <code>./email</code> modules; tRPC caller built with a test context helper
Live-database query tests	SQL emitted by Drizzle against TiDB	Helpers imported from the real <code>./db</code> module and executed against the development database
Type checking	Whole codebase	<code>tsc --noEmit</code> with strict typing via tRPC shared contracts
Visual verification	Rendered pages	Dev-server screenshots of public pages and all nine admin pages
Manual acceptance	End-to-end flows	Authenticated browser sessions for booking, ticket email, check-in, and settings flows

Permission tests deliberately exercised negative cases (non-admins blocked from admin endpoints, owners blocked from demoting themselves) in addition to positive cases. The email

tests use a mocked fetch against the Resend API, so no real messages were sent during automated runs; a separate validation test confirms the Resend API key authenticates.

3. Test Inventory

The suite comprises **six test files and 77 test cases**, all passing.

File	Cases	Coverage
<code>server/auth.logout.test.ts</code>	1	Session logout invalidates the auth session
<code>server/booking.test.ts</code>	46	Core booking lifecycle, cost calculation, unique references, multi-day bookings, tour slots, facets, itinerary export/share, user management permissions, audit events, and the analytics date-range regression suite
<code>server/create-email.test.ts</code>	13	Booking confirmation email trigger, fallback to account email, email-provider failure resilience, status-change emails (confirmed/cancelled, no email on unchanged status), and capacityCheck semantics
<code>server/email.test.ts</code>	1	Resend API key validation
<code>server/enhancements.test.ts</code>	11	Email helper behaviour (mocked provider), CSV export format and date-range filtering, and the 14-day daily forecast computation
<code>server/round7.test.ts</code>	9	Site settings round-trip and fallback defaults, gate check-in and undo, cancelled-booking rejection, unknown reference handling, and user creation with duplicate rejection

4. Representative Results

The full output of the suite at the final checkpoint is reproduced below.

```
Test Files  6 passed (6)
  Tests    77 passed (77)
Duration 3.1s
```

Representative verified behaviours include: the system generates a unique `KNMP-XXXXXX` reference per booking; the total cost updates as category quantities change; non-admin users receive `FORBIDDEN` errors on administrative endpoints; audit events are written for role changes, account deactivation, content edits, status changes, gate check-ins, and user creation; the category-breakdown CSV carries a header and one row per category honouring the applied range; the forecast returns exactly fourteen rows covering today through today plus thirteen days; and check-ins are refused for cancelled bookings.

5. Defects Found and Resolved

One production-grade defect was discovered during acceptance of the date-range analytics feature. The monthly-trends and category-breakdown queries failed with a generic “Failed query” error whenever a visit-date range was supplied, even though the equivalent SQL executed correctly in direct database sessions.

The root cause was identified through incremental reproduction scripts: TiDB’s prepared-statement path (used by the `mysql2` driver underlying Drizzle) rejects `SELECT`-list aliases referenced in `ORDER BY`. Both trend queries ordered by aliases (`ORDER BY month ASC` , `ORDER BY revenuePesewas DESC`). The fix replaced alias-based ordering with positional ordering (`ORDER BY 1 ASC` , `ORDER BY 4 DESC`) in `getMonthlyTrends` and `getCategoryBreakdown`. Three live-database regression tests were added to prevent recurrence, and the analytics page was verified visually with ranges applied.

A second, minor defect found during the Round 7 build (a transient duplicate module export and a hoisted-module import resolution) was corrected before checkpointing; neither reached a deployed build.

6. Verification Summary

Area	Verification	Outcome
Public site (Home, Explore, Book, My Bookings, Itinerary)	Screenshots + logged-in manual flows	Render and function correctly on desktop and mobile
Admin dashboard (all nine pages)	Screenshots + authenticated flows	Render and function correctly
PDF tickets with QR codes	pdfkit buffer tests	PDFs render with scannable QR encoding KNMP-TICKET :<reference>
Email delivery	Mocked-provider tests + API key validation	Confirmation and status emails constructed and dispatched
Gate check-in	Live query tests (lookup, check-in, undo, cancellation guard)	Correct behaviour, audit events recorded
Analytics date-range	Regression tests + screenshots	Charts and CSV export respect the range
Capacity warnings	capacityCheck tests against live database	Correct near/over-capacity flags using configured settings

7. Known Gaps and Recommendations

Automated coverage currently sits at the backend layer; the frontend UI is verified through screenshots and manual flows rather than browser automation. Recommended follow-ups include component tests for the booking-form capacity banner logic, Playwright-based end-to-end flows for the booking-to-ticket-email loop, and load testing of the analytics queries at higher booking volumes.